Zijin Zhang

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Positions

University of Alberta, Department of Physics, Postdoctoral Fellow — Sept 2026 – present
- Supervisor: Prof. *Yangyang Shen*

Education

Ph.D. **University of California, Los Angeles (UCLA)**, Ph.D. in Planetary Science — 2022 – 2026
- Thesis: Kinetic-scale solar wind current sheets: statistical characteristics and their role in energetic particle transport
- Advisors: Prof. *Vassilis Angelopoulos* and Prof. *Anton Artemyev*

B.Sc. **University of Science and Technology of China (USTC)**, B.Sc. in Space Physics — 2018 – 2022
- Thesis: Kinetic simulation of the interaction between the Moon's magnetic anomalies and the solar wind (DOI: 10.13140/RG.2.2.15841.68968)
- Advisor: Prof. *Xin Tao*

University Corporation for Atmospheric Research, NASA's Living with a Star Heliophysics Summer School — 2024

Research Interests

- Magnetosphere-ionosphere coupling: energetic particle precipitation and wave-particle interactions
- Heliophysics: Solar wind current sheets and energetic particle transport
- Computational plasma physics and space physics environment data analysis

Publications

The JuliaSpacePhysics ecosystem: High-performance, modular, and scalable software for space physics — In preparation
Zhang, Z., Angelopoulos, V, et al.

Energetic particle transport driven by solar wind current sheets — In preparation
Zhang, Z., Artemyev, A. V., Angelopoulos, V.

Separating Field-Line Curvature and Wave Scattering in Nightside Energetic Electron Precipitation With ELFIN — In preparation
Zhang, Z., Shen, Y., Artemyev, A. V., Angelopoulos, V.

High-Frequency EMIC Wave Power Inferred From Relativistic Electron Precipitation: An ELFIN–PWING Conjunction Survey — Submitted
Zhang, Z., Shiokawa, K., Artemyev, A. V., Mourenas, D., Angelopoulos, V.

An Empirical Electron Precipitation Model During Magnetospheric Substorms — Submitted
CRediT contributions: Resources; Data Curation; Writing – Review and Editing.
Sun, W., Zhang, X.-J., Artemyev, A. V., Kamaletdinov, S., Liang, J., **Zhang, Z.**, Angelopoulos, V.
(Journal of Geophysical Research: Space Physics)

Properties of current sheets in two-dimensional tearing-mediated magnetohydrodynamic turbulence — 2026
Shi, C., Velli, M., Sioulas, N., **Zhang, Z.**
10.3847/1538-4357/ae742a (Astrophysical Journal)

Peculiarities of precipitating electron spectra revealed through a combined DMSP-ELFIN data set — 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Zhang, X.-J.
10.1029/2026JA035299 (Journal of Geophysical Research - Space Physics)

Multifluid equilibrium model of current sheets with interpenetrating ion beams 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Velli, M.
10.1063/5.0331297 (Physics of Plasmas)

Comparison of solar wind current sheets in the inner heliosphere 2026
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Shi, C., Phan, T.
10.3847/1538-4357/ae46a1 (Astrophysical Journal)

Quantification of ion scattering by solar-wind current sheets: Pitch-angle diffusion rates 2025
Zhang, Z., Artemyev, A. V., Angelopoulos, V.
10.1103/pkzv-k5t3 (Physical Review E)

Solar wind discontinuities in the outer heliosphere: Spatial distribution between 1 and 5 AU 2025
Zhang, Z., Artemyev, A. V., Angelopoulos, V., Vasko, I.
10.1029/2025JA034039 (Journal of Geophysical Research - Space Physics)

Relativistic electron flux decay and recovery: Relative roles of EMIC waves, chorus waves, and electron injections 2024
Zhang, Z., Artemyev, A., Mourenas, D., Angelopoulos, V., Zhang, X.-J., et al.
10.1029/2024JA033174 (Journal of Geophysical Research - Space Physics)

A search for technosignatures around 11680 stars with the Green Bank Telescope at 1.15–1.73 GHz 2023
Margot, J.-L., Li, M. G., Pinchuk, P., … **Zhang, Z.**
10.3847/1538-3881/acfda4 (Astronomical Journal)

Awards

- Betty and S.L. Huang graduate fellowship (2026)
- Outstanding student scholarship, USTC (2022)
- Outstanding student scholarship, USTC (2021)
- Social practice (field investigation in natural sciences) individual & team awards (2021)
- Outstanding student scholarship, USTC (2019)

Software

SPEDAS.jl: Julia-based space physics environment data analysis software (DOI: 10.5281/zenodo.15181866)

PlasmaBO.jl: Efficient plasma electromagnetic dispersion-relation solver (DOI: 10.5281/zenodo.18058843)

PlanetaryMagneticFields.jl: A unified framework for planetary magnetic field modeling (DOI: 10.5281/zenodo.18428922)

Other Experience

Artificial Intelligence of Things Lab, Undergraduate Research Assistant 2021 – 2022
- Implemented a distributed system to monitor edge devices and automate IT deployment and management

University of California, Los Angeles, Teaching Assistant
- EPS SCI M7: Perils of Space — Introduction to Space Weather (2025)
- EPS SCI 9: Solar System and Planets (2025)

University of Science and Technology of China
- Teaching Assistant, Optics and Thermodynamics (2022)
- Student President, USTC Student Scientific Expedition Association (2021-2022)